

# Darius Ardales

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## EXPERIENCE

### Data Scientist

*UnionBank of the Philippines*

Nov 2025 – Present

*Pasig, NCR, Philippines*

- Generated multi-million peso quarterly incremental sales by migrating and optimizing a high-revenue forecasting model

### Data Science and AI Intern

*Bank of the Philippine Islands (BPI)*

May 2025 – Aug 2025

*Makati, NCR, Philippines*

- Deployed a prioritization model via Microsoft Copilot Studio, accelerating DS team's project selection by projected 40%
- Built a baseline audio-to-insights pipeline on GCP/Azure serving as foundation for unstructured audio data transformation
- Presented weekly model recommendations to the Head of Data Science, enabling strategic prioritization of model exploration

### Freelance AI Writing Evaluator and Trainer

*Outlier*

Aug 2024 – May 2025

*San Francisco, CA, USA*

- Provided human feedback for Meta's RLHF pipeline, evaluating 80+ Meta AI outputs weekly to boost response accuracy
- Ranked Senior Reviewer and inducted into Oracle top contributors group for consistent annotation quality

### Software Engineer Intern

*Thinking Machines Data Science, Inc.*

Jun 2024 – Aug 2024

*Taguig, NCR, Philippines*

- Led a project from design to presentation, managing stakeholders and delivering satisfied proposal outcomes
- Enhanced office space utilization and projected 60% cost reduction with a web app alternative to HotDesk

## AWARDS

### 1st Place Winner, BPI DATA Wave 2025: Hyper-Personalization & Customer Experience Track

Sep 2025

- Architected a proactive solution projected to slash cost-per-case by 31% and double the customer NPS growth
- Engineered an agentic tool using Google ADK and A2A for automated issue resolution and pre-analyzed agent reporting

### Winner, Maya Big 3 Startup Showdown

Apr 2025

- Developed an AI workflow and predictive models for cashflow/credit analysis with projected 100% loan decision speedup
- Designed and optimized complex SQL queries to extract critical data from our strategic partner's proprietary database

## PUBLICATIONS

### Investigating the Generalization of Transformer Models for Time Series Forecasting in Sinusoidal Functions (Bachelor's Thesis)

Aug 2025

*DLSU Dr. Andrew L. Tan Data Science Institute (ALTDSI)*

- Achieved 87% lower MSE than Vanilla Transformer and outperformed Informer/Autoformer on unseen frequencies
- Developed Takens embedding and simplex projection methods, improving  $R^2$  from -1.26 to 0.72 on varied distributions

### SentiMetry: A Development of Emotional Wellness Web Application Using AI-Driven Sentiment Analysis

Nov 2024

*Institute of Electrical and Electronics Engineers (IEEE)*

- Combined EmoRoBERTa, Logistic Regression, and Bi-LSTM to predict top-3 emotions across 28 categories
- Generated personalized LLM feedback using Mistral API, with 81% of users finding it empathetic

### Towards a Memory-Efficient Filipino Sign Language Recognition Model for Low-Resource Devices

May 2024

*Computing Society of the Philippines (CSP)*

- Achieved 91% model-size reduction (11.7 MB to 1.0 MB) via post-quantization with only 6.25% accuracy drop
- Outperformed the accuracy of raw-frame CNN (by 74.51%) and LSTM (by 68.75%) by using MediaPipe landmarks

## EDUCATION

### De La Salle University

*B.S. in Computer Science, Major in Software Technology, Minor in Data Science*

Manila, NCR, Philippines

Jul 2020 – Aug 2025

## PROJECTS

### PageLeaper

Jun 2025 – Jul 2025

- Built a RAG system with Gemini embeddings processing 5+ PDFs/DOCX simultaneously with 20-chunk batch optimization

### Google Gen AI Intensive Course Capstone: Central Bank Compass AI

Mar 2025 – Apr 2025

- Engineered a multi-agent Agentic RAG system to distill central bank communications into actionable insights

### Towards a Safer Construction Environment: A Simple CNN for Safety Classification

Nov 2024 – Dec 2024

- Built an end-to-end safety classification pipeline with data augmentation (5x expansion), TTA, and FastAPI deployment

### Harvard Online: CS50's Introduction to Artificial Intelligence with Python

Apr 2024 – Jun 2024

- Gained robust theoretical and mathematical understanding of core AI algorithms from search to neural architectures

### Klima Insights

Mar 2024 – Apr 2024

- Designed a dashboard for NGOs to monitor the impact of global warming in the Philippines using GeoPandas and Plotly

## TECHNICAL SKILLS

**Languages:** Python, SQL, TypeScript, JavaScript

**AI/ML:** Google ADK, LangChain, LangGraph, PyTorch, TensorFlow, scikit-learn, XGBoost, Nixtla, OpenCV, Next.js

**Data:** PostgreSQL, MySQL, Snowflake, ChromaDB, Pandas, NumPy, Matplotlib

**MLOps:** AWS SageMaker, Vertex AI, AWS, GCP, Azure, Git, GitHub, Docker, W&B